

-
1. Let f be a function from S to T and let $X \subset S, Y \subset T$. If $f^{-1}(Y)$ is the inverse image of Y under f , then which of the following statements are true.
- (i) $f(f^{-1}(Y)) \subset Y$ (ii) $f^{-1}(f(X)) \subset X$
- A) only (i) is true B) only (ii) is true
- C) both (i) and (ii) are true D) neither (i) nor (ii) is true
2. Suppose that $\{x_n\}$ is a sequence of real numbers satisfying the following condition. For every $\epsilon > 0$, there exists n_0 such that $|x_{n+1} - x_n| < \epsilon, \forall n \geq n_0$.
The sequence $\{x_n\}$ is
- A) bounded but not necessarily Cauchy
- B) Cauchy but not necessarily bounded
- C) convergent
- D) not necessarily bounded
3. Define $f(x) = \frac{1}{\sqrt{x}}$ for $x > 0$. Then $f(x)$ is uniformly continuous
- A) on $(0, \infty)$
- B) on $[r, \infty)$ for any $r > 0$
- C) on $(0, r]$ for any $r > 0$
- D) only on intervals of the form $[a, b]$ for $0 < a < b < \infty$.
4. Which of the following statements is "not" true?
- A) all continuous and bounded functions are Riemann integrable
- B) all bounded functions which are monotone are Riemann integrable
- C) continuity is a necessary condition for Riemann integrability
- D) boundedness is a necessary condition for Riemann integrability.
5. Which of the following statements is true?
- (i) every Cauchy sequence in a metric space is convergent
- (ii) a closed and bounded metric space is complete
- A) only (i) is true B) only (ii) is true
- C) both (i) and (ii) are true D) neither (i) nor (ii) is true.

6. Consider the metric space $S = [0, 1] \cup \{2\}$ which is a subspace of $\mathbb{R}$.
What can you say about the following statements?
- (i) $[0, 1]$ is an open subset of S
(ii) $\{2\}$ is an open subset of S
- A) only (i) is true B) only (ii) is true
C) both (i) and (ii) are true D) neither (i) nor (ii) is true.
7. Which of the following metric subspaces of $\mathbb{R}$ is "not" complete?
- A) $[0, 1]$ B) $(0, 1)$ C) $[0, 1] \cup [2, 3]$ D) $\{0, 1\}$
8. Which of the following can be linear subspace(s) of $\mathbb{R}^3$?
- (i) $\{x \in \mathbb{R}^3 : x = (x_1, x_2, 0); x_1, x_2 \in \mathbb{R}\}$
(ii) $\{x \in \mathbb{R}^3 : x = (x_1, x_2, 1); x_1, x_2 \in \mathbb{R}\}$
- A) only (i) B) only (ii) C) both (i) and (ii) D) neither (i) nor (ii)
9. The vectors $X = [3, -9, 12]$ and $Y = [-4, 12, -16]$ are such that
- A) $4X - 3Y = 0$ B) $4X + 3Y = 0$
C) $3X + Y = 0$ D) X and Y are linearly independent
10. For which of the following, $B = (v_1, v_2)$ is "not" a basis of $\mathbb{R}^2$?
- A) $v_1 = (1, 1), v_2 = (1, -1)$ B) $v_1 = (0, 1), v_2 = (1, 0)$
C) $v_1 = (1, -1), v_2 = (-1, 1)$ D) $v_1 = (0, 1), v_2 = (1, -1)$
11. Which of the following statements is "not" true?
- A) the trace of an idempotent matrix is always an integer
B) all the eigen values of an idempotent matrix are equal to 1
C) identity matrix is an idempotent matrix
D) all idempotent matrices other than identity matrix are singular
12. The eigen values of the matrix $\begin{bmatrix} 1 & 2 \\ 5 & 4 \end{bmatrix}$ are
- A) 6 and -1 B) -6 and 1 C) -6 and -1 D) 6 and 1
13. Which of the following matrices is "not" positive definite?
- A) $\begin{bmatrix} 1 & 2 \\ -4 & 5 \end{bmatrix}$ B) $\begin{bmatrix} 1 & -2 \\ 5 & -4 \end{bmatrix}$ C) $\begin{bmatrix} 6 & 4 \\ 5 & 4 \end{bmatrix}$ D) $\begin{bmatrix} 1 & 2 \\ 4 & 5 \end{bmatrix}$

14. If $A = \{1, 2, 3, 4, 5, 6\}$, which of the following statements is/are true?
 (i) $\{\{\}, \{2, 3\}, \{1, 4\}, \{5, 6\}, A\}$ is a sigmafield of subsets of A
 (ii) $\{A, \{\}\}$ is a sigmafield of subsets of A
 A) only (i) is true B) only (ii) is true
 C) both (i) and (ii) are true D) neither (i) nor (ii) is true.
15. Consider the sequence $\{A_n\}$, where $A_n = \{w : 0 < w < 1 + (-1)^n/n\}$. Then which of the following statements is "not" true
 A) $\overline{\lim} A_n = \{w : 0 < w < 1\}$ B) $\overline{\lim} A_n = \{w : 0 < w \leq 1\}$
 C) $\underline{\lim} A_n = \{w : 0 < w < 1\}$ D) $\lim A_n$ does not exist
16. In a class, there are 5 boys and 10 girls. If 6 students are selected at random, what is the probability that a particular boy gets selected?
 A) 1/6 B) 5/15 C) 3/5 D) 6/15
17. If the events A and B have probabilities $P(A)=0.3$ and $P(B)=0$, then
 (i) A and B are mutually exclusive
 (ii) A and B are independent
 (iii) A and B are mutually exhaustive
 A) Only (i) is true B) only (i) and (ii) are true
 C) (i), (ii) and (iii) are true D) only (ii) and (iii) are true
18. What can you say about the following statements?
 (i) If $P(A)=0$, A is an impossible event
 (ii) If A and B are independent non-null events, they cannot be disjoint
 A) only (i) is true B) only (ii) is true
 C) both (i) and (ii) are true D) neither (i) nor (ii) is true.
19. If there are 53 Sundays in a randomly selected year, what is the probability that it is a leap year?
 A) 1/3 B) 2/7 C) 2/5 D) 3/5
20. For a multiple choice question, the students have to select the correct choice among 4 choices given. If two students make random guesses independently, what is the probability that both of them make wrong choices but their choices are same?
 A) 9/16 B) 3/16 C) 1/16 D) 6/16
21. For a multiple choice question, the students have to select the correct choice among 4 choices given. If two students make random guesses independently, what is the probability that both of them make wrong choices given their choices are different?
 A) 1/2 B) 3/8 C) 3/16 D) 1/4

22. Which of the following statements are true about the pdf $f(x)$ of a continuous random variable X . ?
- (i) $0 \leq f(x) \leq 1$ (ii) $f(x) = P(X=x)$
- A) only (i) is true B) only (ii) is true
- C) both (i) and (ii) are true D) neither (i) nor (ii) is true.
23. Which of the following are properties of $\bar{F}(x) = P(X > x)$ where X is a discrete random variable?
- (i) $\bar{F}(x)$ is non-increasing (ii) $\bar{F}(x)$ is continuous from the left
- A) only (i) is true B) only (ii) is true C) both (i) and (ii) are true D) neither (i) nor (ii) is true.
24. Which of the following is a bivariate joint pdf?
- (i) $f(x, y) = \frac{e^{-x}}{(y+1)^2}, 0 < x < \infty, 0 < y < \infty$ and $f(x, y) = 0$; otherwise
- (ii) $f(x, y) = \frac{1}{(xy)^2}, 1 < x < \infty, 1 < y < \infty$ and $f(x, y) = 0$; otherwise
- A) only (i) is true B) only (ii) is true C) both (i) and (ii) are true D) neither (i) nor (ii) is true
25. Which of the following is "not" true for a non-negative random variable X , assuming that the expectations exist?
- A) $E(X^3) \geq [E(X)]^3$ B) $E(1/X) \leq 1/E(X)$
- C) $E(e^X) \geq e^{E(X)}$ D) $E[(X-1)^2] \geq [E(X-1)]^2$
26. Which of the following "cannot" be true, if μ_r denote the r^{th} central moment of a continuous random variable?
- A) $\mu_2 = 9, \mu_3 = -6, \mu_4 = 90$ B) $\mu_2 = 8, \mu_3 = 6, \mu_4 = 64$
- C) $\mu_2 = 7, \mu_3 = -6, \mu_4 = 50$ D) $\mu_2 = 6, \mu_3 = 6, \mu_4 = 90$
27. If a random variable has moment generating function $M_X(t) = (1 + e^t)^8/2^8$, the variance of the random variable is
- A) 8 B) 4 C) 2 D) 1
28. If $\{X_n\}$ is a sequence of random variables defined as $X_n = 0$ if no head is obtained in the first n tosses and $X_n = 1$ if atleast one head is obtained in the first n tosses of an unbiased coin, then which of the following statements is/are true?
- (i) $\{X_n\}$ converges in probability to 1. (ii) converges almost surely to 1.
- A) only (i) is true B) only (ii) is true C) both (i) and (ii) are true D) neither (i) nor (ii) is true
29. The mode of convergence stated in central limit theorem is
- A) weak convergence B) convergence in probability C) almost sure convergence D) none of these
30. If $X_1, X_2, \dots, X_n$ are independent and identically distributed geometric random variables, then their sum follows — distribution.
- A) Binomial B) Negative binomial C) Geometric D) Hypergeometric
31. In a random experiment, a coin is tossed until a head is obtained and the experiment is repeated 100 times. If the average number of tails obtained before the first head is 2, probability of obtaining head can be estimated using method of moments as
- A) 1/2 B) 1/3 C) 2/3 D) 1/50

32. The mode of a Binomial distribution with parameters n and p can be zero when
 A) $n \geq 1/p$ B) $n \geq (1-p)/p$ C) $n \leq 1/p$ D) $n \leq (1-p)/p$
33. Buses arrive at a bus stop at 30 minute intervals starting from 7 am. If a person arrives at a random time between 8 and 9 am, his waiting time follows
 A) exponential distribution with mean 15 minutes B) exponential distribution with mean 30 minutes
 C) uniform distribution with mean 15 minutes D) uniform distribution with mean 30 minutes
34. A long-distance calling plan charges a flat rate of 99 paise for calls up to 20 minutes in duration and then 10 paise for each additional minute exceeding 20 (assume that calls lasting a non-integer number of minutes are charged proportionately to a whole minute's charge). What will be the expected cost of the plan, if the distribution of consumer's call duration is exponential with expected call duration 10 minutes ?
 A) $99 + 100e^{-2}$ B) $99 + 100e^{-10}$ C) $99 + 100e^{-0.1}$ D) $99 + 100e^{-200}$
35. Which of the following statements is/are true?
 (i) If X follows Normal distribution, $\log(X)$ follows lognormal distribution
 (ii) If X follows Exponential distribution, X^2 follows Weibull distribution
 A) only (i) is true B) only (ii) is true C) both (i) and (ii) are true D) neither (i) nor (ii) is true
36. If $X_1, X_2, \dots, X_{10}$ are independent and identically distributed Exponential random variables with mean 1, the distribution of $(X_1 + X_2 + X_3 + X_4)/(X_5 + X_6 + \dots + X_{10})$ is
 A) Type I Beta B) Type II Beta C) Gamma D) None of these
37. The r^{th} order statistic of a sample of size n from Uniform(0,1) population is distributed as Beta Type I with parameters
 A) r and $n-r$ B) r and $n-r+1$ C) $r-1$ and $n-r$ D) $r-1$ and $n-r+1$
38. Which of the following statements is/are true?
 (i) If X follows standard Normal distribution, X^2 follows Gamma distribution
 (ii) If X_1 and X_2 are independent Exponential random variables with the same mean, $X_1 + X_2$ follows Gamma distribution
 A) only (i) is true B) only (ii) is true C) both (i) and (ii) are true D) neither (i) nor (ii) is true
39. If $\bar{X}$ is the sample mean and Y is the sum of squares of deviations of the observations from the sample mean when sampling from a Normal population with mean μ , then $n(n-1)(\bar{X} - \mu)^2/Y$ is distributed as
 A) Student's t with $n(n-1)$ df B) F distribution with 1 and $n(n-1)$ df
 C) F distribution with 1 and $n-1$ df D) None of these
40. For a random sample of size 11 from a normal population, the sum of observations is 44 and sum of squares of observations is 220. Then an unbiased estimator of the population variance gives the estimate
 A) 4 B) 4.4 C) 6 D) 6.6
41. If F_{n_1, n_2} denote an F statistic with (n_1, n_2) degrees of freedom, which of the following is true?
 A) $P(F_{n_1, n_2} > k) = 1/P(F_{n_2, n_1} < k)$ B) $P(F_{n_1, n_2} > k) = P(F_{n_2, n_1} > 1/k)$
 C) $P(F_{n_1, n_2} > k) = P(F_{n_2, n_1} < 1/k)$ D) None of these

42. If $Y_1, Y_2, \dots, Y_6$ is a random sample of observations from a Normal population with zero mean, then what can you say about the expected value of $(Y_1^2 + Y_2^2 + Y_3^2)/(Y_4^2 + Y_5^2 + Y_6^2)$?
- A) 0 B) 1 C) 3 D) Expectation does not exist.

43. The joint pdf of random variables X and Y is given by

$$f_{X,Y}(x, y) = \begin{cases} 8xy, 0 < x < y < 1 \\ 0, \text{otherwise} \end{cases}$$

Then expected value of X is

- A) 8 B) 8/15 C) 4/5 D) None of the above
44. The joint pdf of random variables X and Y is given by

$$f_{X,Y}(x, y) = \begin{cases} e^{-(x+y)}, 0 < x < \infty, 0 < y < \infty \\ 0, \text{otherwise} \end{cases}$$

Then coefficient of correlation between X and Y is

- A) 0 B) 1/2 C) -1/2 D) 1/4
45. Which of the following statements is/are true?
- If X and Y are independent, the slope of the regression line of Y on X is zero.
 - If both the regression coefficients (corresponding to X on Y as well as Y on X) are zero, the variables X and Y are independent
- A) i. only B) ii only C) both i and ii D) neither i nor ii

46. Suppose k is the mean square deviation from mean for a population of size 1000. The sampling variance of the mean for sample size 10, in sampling without replacement is
- A) $99k/999$ B) $99k/1000$ C) $99k/990$ D) $999k/1000$

47. In sampling with replacement from a population of size 2000 units, the number of equally likely samples of size 10 is
- A) $\frac{2000!}{10! 1990!}$ B) $\frac{2000!}{1990!}$ C) 2000^{10} D) 10^{2000}

48. A sample of size 5 is to be taken from a population of size 999 with the units numbered randomly from 1 to 999. If sample is selected by circular systematic sampling, which of the following statements is/are not true?
- Sample mean is not an unbiased estimate of the population mean
 - The samples of the required sample size can be obtained
 - It is not possible to obtain an unbiased estimate of the sampling variance of the estimate of population mean using a single sample
 - All the above statements are not true

49. If s^2 is the mean square deviation of observations from sample mean when a random sample of size n is taken with replacement from a Normal population, what can you say about the following statements?
- (i) s^2 is a consistent but biased estimator of the population variance
(ii) $ns^2/(n-1)$ is a consistent and unbiased estimator of the population variance
- A) only (i) is true B) only (ii) is true C) both (i) and (ii) are true D) neither (i) nor (ii) is true
50. A researcher wishes to estimate the mean height of a population of trees. Suppose a sample of trees has mean height 100 inches and mean diameter 10 inches. If it is known that the mean diameter of the population of trees is 11 inches, a ratio estimate of the population mean height is
- A) 1000/11 inches B) 110 inches C) 100 inches D) None of these
51. The following data shows the number of defectives per packet in 100 packets of 10 units each.
- | | | | | | | |
|--------------------------------|----|----|----|----|---|---|
| No. of defectives per packet : | 0 | 1 | 2 | 3 | 4 | 5 |
| No. of packets : | 22 | 36 | 28 | 11 | 2 | 1 |
- An estimate of the proportion of defective units is
- A) 138/1000 B) 15/100 C) 133/1000 D) 14/100
52. In a sample of 1000 men, 350 were unemployed and in a sample of 800 women, 250 were unemployed. Estimate of the proportion of employed persons in the population is
- A) 1/3 B) 2/3 C) $(65/100 + 55/80)/2$ D) None of these
53. In two independent samples of size 20 and 30 taken from a Normal population, the mean square deviations from sample mean are 21 and 18 respectively. An unbiased estimate of the population variance is
- A) 20 B) 19.2 C) 19.6 D) 19
54. To test the null hypothesis that mean weight of adolescent female population in India is 50 kg against the alternative that it is greater than 50 kg, it is decided to reject the null hypothesis if mean of a sample of size n is greater than 55 kg. Assuming that the population is normally distributed with standard deviation 10 kg, what should be the minimum sample size so that maximum probability of type I error is 0.025?
- A) 4 B) 8 C) 16 D) 22
55. In testing the null hypothesis that mean of an exponential population is 2 against the alternative that it is 4, it is decided to take an observation and reject the null hypothesis if its value is greater than 3. What is the power of the test?
- A) $e^{-3/4}$ B) e^{-12} C) $1 - e^{-3/4}$ D) $1 - e^{-12}$
56. A non-parametric test for testing whether a given sample is from a population with a specified continuous probability distribution.
- A) Kruskal-Wallis test B) Mann-Whitney U test C) Wald-Wolfowitz run test D) Kolmogorov-Smirnov test

57. Let one observation corresponding to i^{th} block and j^{th} treatment be missing in an RBD and let G' denote the grand total of known observations and B' and T' denote the total of known observations in the i^{th} block and j^{th} treatment respectively. If there are b blocks and t treatments, the missing observation can be estimated as
- A) $(bB' + tT' - G')/((b-1)(t-1))$ B) $(tB' + bT' - G')/((b-1)(t-1))$
C) $(bB' + tT' - G')/(bt-1)$ D) $(tB' + bT' - G')/(bt-1)$
58. Which of the following statements is/are true?
- (i) The Kruskal-Wallis test is used for comparing two or more independent samples of equal or different sample sizes
(ii) The Friedman test is a nonparametric test used to compare three or more related groups to see if they are significantly different
- A) only (i) is true B) only (ii) is true C) both (i) and (ii) are true D) neither (i) nor (ii) is true
59. If the 95% confidence interval for the mean μ of a normal population is $(0.6, 3.6)$, which of the following statements is true?
- (i) We can reject the hypothesis that $\mu = 0$ against the alternative $\mu \neq 0$ at 5% level of significance
(ii) We can accept the hypothesis that $\mu = 1$ against the alternative $\mu \neq 1$ at 5% level of significance
- A) only (i) is true B) only (ii) is true C) both (i) and (ii) are true D) neither (i) nor (ii) is true
60. Suppose $X_1, X_2, \dots, X_n$ are independent and identically distributed Poisson random variables with parameter λ , where λ has an exponential distribution with mean 1. Under squared-error loss function, the estimate of λ is
- A) $\sum_{i=1}^n X_i/n$ B) $(\sum_{i=1}^n X_i + 1)/n$ C) $\sum_{i=1}^n X_i/(n+1)$ D) $(\sum_{i=1}^n X_i + 1)/(n+1)$
61. Which of the following statements is "not" true?
- A) The Bayes estimator under quadratic error loss function is the prior mean.
B) The Bayes estimator under absolute error loss function is the posterior median.
C) The Bayes estimator minimises the expected posterior loss.
D) The Beta family is a conjugate family of prior distributions for Bernoulli samples
62. Let $\underline{X}$ be a random vector following multivariate normal distribution $N_p(\mu, \Sigma)$, where Σ is non-singular. Let $\underline{X}$ be partitioned as $\underline{X} = [X^{(1)} X^{(2)}]'$ and $\underline{\mu} = [\mu^{(1)} \mu^{(2)}]'$ be the corresponding partition of $\underline{\mu}$. Let $\Sigma_{rs} = E[(X^{(r)} - \mu^{(r)})(X^{(s)} - \mu^{(s)})']$; $r = 1, 2$; $s = 1, 2$. Then $X^{(1)} + BX^{(2)}$ is independent of $X^{(2)}$ if B is
- A) $-\Sigma_{12}\Sigma_{22}^{-1}$ B) $\Sigma_{12}\Sigma_{22}^{-1}$ C) $\Sigma_{11} + \Sigma_{12}\Sigma_{22}^{-1}$ D) $\Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}$

63. Let $\underline{Y}$ be a random vector following multivariate normal distribution $N_p(\mu, I)$ and let $\underline{Y}'\underline{Y} = \underline{Y}'A_1\underline{Y} + \underline{Y}'A_2\underline{Y}$ where A_1 and A_2 are idempotent. Then which of the following statements are true.
- (i) $\underline{Y}'A_1\underline{Y}$ and $\underline{Y}'A_2\underline{Y}$ follow chi-square distribution.
(ii) $\underline{Y}'A_1\underline{Y}$ and $\underline{Y}'A_2\underline{Y}$ are independent.
- A) only (i) is true B) only (ii) is true C) both (i) and (ii) are true D) neither (i) nor (ii) is true
64. Let $\underline{Y}_1, \underline{Y}_2, \dots, \underline{Y}_n$ be a random sample from $N_p(\mu, \Sigma)$ and let S be the sample dispersion matrix which is unbiased estimator of Σ . The sampling distribution of S is
- A) Wishart with parameters Σ and n B) Wishart with parameters Σ and $n - 1$
C) Wishart with parameters $\frac{1}{n}\Sigma$ and $n - 1$ D) Wishart with parameters $\frac{1}{n-1}\Sigma$ and $n - 1$
65. If $T^2 = nY'A^{-1}Y$ is distributed as Hotelling's T^2 with n degrees of freedom. Then which of the following statements can be true.
- (i) Y follows $N_p(0, I)$ and A follows Wishart with parameters I and n .
(ii) Y follows $N_p(0, \Sigma)$ and A follows Wishart with parameters Σ and n .
- A) only (i) is true B) only (ii) is true C) both (i) and (ii) are true D) neither (i) nor (ii) is true
66. Which of the following statements are true about the sample multiple correlation coefficient R when the corresponding population correlation coefficient is zero.
- (i) $R^2/(1 - R^2)$ is distributed as the ratio of independent chi-square.
(ii) R^2 follows chi-square distribution.
- A) only (i) is true B) only (ii) is true C) both (i) and (ii) are true D) neither (i) nor (ii) is true
67. Let $\bar{X}$ and A be the maximum likelihood estimators of μ and Σ based on a random sample from multivariate normal population $N_p(\mu, \Sigma)$. Then
- (i) $\bar{X}$ and A are independently distributed.
(ii) $\bar{X}$ and A are consistent estimators of μ and Σ .
- A) only (i) is true B) only (ii) is true C) both (i) and (ii) are true D) neither (i) nor (ii) is true
68. Which of the following statements is/are true?
- (i) In a Markov chain with a finite number of states, not all of the states can be transient
(ii) All states in a finite irreducible Markov chain are recurrent
- A) only (i) is true B) only (ii) is true C) both (i) and (ii) are true D) neither (i) nor (ii) is true

69. Which of the following statements is/are true?
- (i) If one state in an equivalent class is transient, then all states are transient
 - (ii) If one state in an equivalent class is recurrent, then all states are recurrent
- A) only (i) is true B) only (ii) is true C) both (i) and (ii) are true D) neither (i) nor (ii) is true
70. In a gambling game, the probability of winning of a particular gambler is 0.6, each time he plays. If he wins, his fortune increases by *Rs.*100 and a loss costs him *Rs.*100. Suppose he starts out with *Rs.*100 and will quit playing as soon as his fortune hits *Rs.*0 or *Rs.*500. What is the probability that he will eventually get *Rs.*500?
- A) 81/211 B) 16/211 C) 27/211 D) 32/211
71. Customers arrive at a window in a bank, according to a Poisson distribution with mean 10 per hour. Service time per customer is exponential with mean 4 minutes. How long is an arriving customer expected to wait before being served?
- A) 4/15 B) 2/15 C) 3/10 D) 1/5
72. In a railway marshalling yard, goods trains arrive at a rate of 36 trains per day. Assuming that the inter-arrival time follows exponential distribution and the service time distribution is also exponential with an average of 30 minutes, find the probability that the queue size exceeds 10.
- A) 0.75^9 B) 0.75^{10} C) 0.75^{11} D) 0.75^{12}
73. Assume that $X(t)$ is a random process defined as follows: $X(t) = A \cos(2\pi t + \phi)$, where A is a zero-mean normal random variable with variance $\sigma_A^2 = 2$ and ϕ is uniformly distributed random variable over the interval $-\pi \leq \psi \leq \pi$. A is statistically independent of t and ϕ . Let the random variable Y be defined as $Y = \int_0^1 X(t) dt$. What will be the mean of Y ?
- A) 0 B) 1 C) π D) $\pi/2$
74. Which of the following is true about the assumptions in Poisson process?
- (i) The process has independent increments
 - (ii) If the system is in a given state, the probability distribution of the time t until the next arrival is exponential
- A) only (i) B) only (ii) C) both (i) and (ii) D) neither (i) nor (ii)
75. Which of the following methods can be used for estimating seasonal components in a time series?
- (i) Ratio to trend method
 - (ii) Moving average method
- A) only (i) B) only (ii) C) both (i) and (ii) D) neither (i) nor (ii)
76. Which of the following index numbers satisfy circular test?
- (i) Simple geometric mean of price relatives
 - (ii) Simple aggregative index
 - (iii) Fisher's ideal index
- A) only (i) and (ii) B) only (iii) C) (i), (ii) and (iii) D) None of these

77. Which of the following index numbers satisfy time reversal test?

(i) Simple geometric mean of price relatives

(ii) Simple aggregative index

(iii) Edgeworth-Marhsall index

A) only (i) and (ii)

B) only (ii) and (iii)

C) (i), (ii) and (iii)

D) None of these

78. Which of the following statements is "not true" ?

A) The set of all rational numbers is neither closed nor open

B) The intersection of any collection of open sets is open

C) The intersection of any collection of closed sets is closed

D) The set of all integers is a closed set

79. If the n^{th} term of a series is $a_n = n(1/2)^n$, the sum of the series $\sum_{n=0}^{\infty} a_n$ is

A) 1

B) 2

C) 4

D) 8

80. Which of the following statements is/are true?

(i) The function $f(x) = |x|$ is not continuous on $(-1, 1)$.

(ii) The function $f(x) = |x|$ is integrable on $(-1, 1)$.

A) only (i)

B) only (ii)

C) both (i) and (ii)

D) neither (i) nor (ii)